Finite Address Component Replay and an SU(3) Singlet Audit

A conditional WRRA computation and a precise limit on address-color coupling

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Abstract

We report a bounded reproducibility study of the particle/component branch of Minimal Computing Cosmology 3.0, distinguishing its arithmetic component outputs from its separately adopted color representation. Frozen-input replay reproduces 427,892 odd-composite addresses, phenotype weight 0.050000000000000065 and assembly-order total-variation distance 0.20472100116369085. The supplied antisymmetric three-color state is annihilated by all eight total generators. Exact singlet support in three fundamental factors forces address-color factorization, restricting what a color-only interface can retain. Source inspection and restricted-namespace execution establish that the original color routine reads no arithmetic state. Together these results locate the interface boundary: order-sensitive component information remains in the address/component register, while the exact three-factor color singlet supplies a consistent but address-independent color sector. The work is a bounded reproducibility and implementation note; it adds no new SU(3) representation theory, particle identification or observational validation.

1. Scope and reproducibility question

The question is whether a successful combined Stage-3 replay supports each proposed claim about its arithmetic component and color interfaces. Component statistics and color invariance are different outputs of the same package. The replay measures an order ambiguity relevant to the component decoder; dependency inspection tests whether the color output depends on that decoder; the kernel proof restricts what the declared color state can encode. Neither branch alone answers all three interface questions.

The contribution is a source-specific reproducibility record and dependency map for researchers inspecting or extending this finite model. Its structural finding is the separation between order-sensitive arithmetic assembly and an address-independent color contract. The original decoder is already labeled conditional: this note documents the existing boundary and does not attribute an unsupported claim or a deployed decoder failure to that source.

Assessment contract

Inputs: frozen address calibration, inherited assembly/charge rules, adopted three-color representation.
WRRA transformation: weighted address assembly and color checks feed a dependency and information-retention audit.
Outputs: reproduced component statistics, color residuals, exact factorization and a source dependency map.
Falsification: failed replay, a violated generator/kernel identity, a counterexample to factorization under the stated support assumptions. An unexpected arithmetic-state read in the isolated color routine would invalidate the dependency conclusion. Observed physical adequacy requires an additional stated mapping to observations.

2. Frozen component replay and the supplied color branch

WRRA abbreviates Worldline-Residue-Resource-Action. The frozen upstream commit is 9d9daf21154e46e55b9fd0fd461c7c070156ab94. Active addresses are n=2,...,1,015,000, with normalized measure w(n)=n^(-alpha)/sum_{m=2}^N m^(-alpha), alpha=1.8996877935161325. The reported phenotype weight is W_phi=sum_n w(n)e_phi(n). The admission threshold h=1.4476744689338703 is retained without refitting. On odd composites, e_phi=1-product_k[1-sigmoid(d_k-h)] for k=0,...,7, where d_k=sum_j cos[gamma_j(log n+0.1k)]/2 and sigmoid(x)=1/(1+exp(-x)). Other addresses have zero phenotype effect. Adopted gamma values are 14.134725141734695, 21.022039638771556, 25.01085758014569 and 30.424876125859512.

```
p(n)=w(n)e_phi(n) / sum_m w(m)e_phi(m)
```

For each pool of the first K prime labels, initialize floor(n/sum(primes)) copies of every prime, then repeatedly cycle through the chosen ascending or descending prime order, taking at most one copy of each prime per traversal when the residue permits, until the residue is below two. Integer reconstruction n=sum_j count_j prime_j+residue is preserved. The candidate fraction is the p-weighted count of historically selected labels {5,47}, divided by the p-weighted total component count. Particle names, those labels and the 16-channel charge table are supplied. The exact rational neutral motifs are uud+electron, udd, an electron/conjugate pair and an individually neutral channel.

Branch	Candidate fraction / mean component count
K15_small	0.129471986937 / 15.9763714572
K15_large	0.127159411697 / 11.4294924701
K16_small	0.124982781217 / 15.2321441711
K16_large	0.111240636923 / 10.5785808287

Define a_j=sum_n p(n)count_j(n) and P_j=a_j/sum_l a_l, aligning prime labels between orders. Then TV=(1/2)sum_j |P_j(small)-P_j(large)|=0.20472100116369085 at K=15. The upstream Stage-3 source already computes this nonzero TV and tests TV>0.1; the present audit reproduces, rather than discovers, that order dependence. Fixed chosen orders remain legitimate algorithms; an order-independent decoder is a stronger claim. No observed neutron abundance is inferred from these conditional component fractions.

```
|S>=(|rgb>+|gbr>+|brg>-|rbg>-|grb>-|bgr>)/sqrt(6)
Q_a=T_a tensor I tensor I + I tensor T_a tensor I + I tensor I tensor T_a
```

Standard fundamental generators satisfy Tr(T_a T_b)=delta_ab/2. The original color_check() uses G_a=2T_a, hence Tr(G_a G_b)=2 delta_ab; annihilation and the kernel are unchanged. It constructs the three-dimensional tensor, signed permutations and eight matrices internally, and reads no address or assembly data.

Input branch	Consumer and consequence
Address measure and admission	Prime assembly; component regression and order sensitivity
Charge table and named motifs	Exact rational neutrality checks
Supplied color states and matrices	Standalone singlet test; not an arithmetic-to-color channel
Generator common kernel	Conditional information-retention constraint

dependency_audit.py recursively checks external bytecode reads and runs the extracted original function with only NumPy, itertools, math and the builtins complex, float, max, range and sum. Those eight names are the entire external-read set; no closure state is captured. This confirms the source-specific dependency finding, assuming ordinary unmodified libraries. Targeted AST extraction bypasses unrelated macro imports; this is not a full Stage-3 pipeline rerun.

3. Exact address-color restriction

Proposition. Let H_A be any finite address space and H_C=(C^3) tensor (C^3) tensor (C^3), carrying the same fundamental SU(3) action on each factor. Let P project onto the common kernel of the eight Q_a. If a density operator Omega satisfies Omega=(I_A tensor P) Omega (I_A tensor P), then P=|S><S| and Omega=sigma_A tensor |S><S| for a unique address density operator sigma_A.

Proof. The two diagonal traceless generators require equal occupation of all three colors, leaving only the six permutations of rgb. For each pair of colors, the raising operator maps two permutations differing by their exchange to the same repeated-color output. Its vanishing forces opposite coefficients. These transposition relations connect all six permutations, fixing their coefficients up to an overall amplitude and permutation parity. Direct substitution verifies the resulting |S> is annihilated by every Q_a, so the kernel has dimension one. Expand Omega=sum_nm |n><m| tensor C_nm. Rank-one support gives C_nm=P C_nm P=c_nm P. Hence Omega=sigma_A tensor P. Positivity and unit trace pass to sigma_A; partial trace gives uniqueness.

For address observable B and color observable C, expectations therefore factor as Tr(sigma_A B) times <S|C|S>. Address-controlled collective transformations U_n tensor U_n tensor U_n with U_n in SU(3) leave |S> fixed and cannot encode different address labels into this sector. A prospective decoder that deletes address labels cannot reconstruct them from this fixed color state. Exact support is stronger than zero mean charge; omitted gluon, flavor or spatial factors can change the applicable representation. Spin, confinement, binding, a particle spectrum and arithmetic selection of SU(3) remain open.

4. Consequence for a particle/component interface

The replay and the color theorem answer complementary questions about the same frozen implementation. Changing the assembly order changes its component-number distribution while preserving integer conservation. The color routine cannot resolve that ambiguity: its function reads no assembly output, and its unique exact singlet carries no independently variable color-state label. Thus the results locate where a later address-to-particle decoder must retain information. They do not establish that such a decoder has already been physically completed.

For example, take two distinct address preparations |a><a| and |b><b|, each tensored with |S><S|. Their address registers are distinguishable, but both reduced color states equal |S><S|. For any color measurement effect E, both give probability <S|E|S>. Deleting the address register therefore makes the preparations indistinguishable by color-only queries. Address-controlled collective SU(3) rotations cannot repair this loss. This example applies the proven restriction to interface design.

A fixed ascending or descending assembly is a valid declared algorithm. The measured TV is a counterexample to equality of the two tested aggregate component distributions; zero TV would not by itself prove equality of per-address outputs or invariance under every possible order. The present calculation therefore rejects only the stronger order-independent interpretation for which this nonzero counterexample is sufficient.

For maintainers extending the Stage-3 decoder, the concrete workflow is: declare which address/component variables the extension reads; retain that register when outputs depend on its labels; rerun integer reconstruction and conditional component comparisons for the declared prime order; and test all eight total generators plus symmetry-breaking controls for any changed color state. The supplied color routine remains an adopted consistency contract until an explicit address-dependent coupling is calculated. This is guidance for using the supplied replay and tests, not a newly implemented coupling or release tool. A larger state space or additional factors may change the information constraint, but no such extension is implemented or claimed here. Reproducing established identities remains a valid consistency result; the new source-specific knowledge is which outputs are actually connected and where the proposed explanatory connection remains open.

5. Verification results and reproduction

Diagnostic	Recorded result
Targeted replay and extended color-audit suite	36 / 36 pass
Source-dependency suite	4 / 4 pass
Common kernel dimension	1, numerical and exact symbolic
Maximum total-generator action	0 in the recorded original run
Generator Gram / commutator errors	1.110e-16 / 1.241e-16
Collective SU(3) rotation error, 32 trials	5.847e-16
Joint-state factorization error	1.507e-16
rgb / one-site phase negative-control charge norms	0.707106781 / 0.816496581

The exact symbolic calculation uses nonzero rescalings of the generators to remove radicals; the common kernel and signed integer singlet vector are checked without numerical tolerance. The joint-state test projects a seeded positive 6-address by 27-color density operator into the exact singlet sector and compares it with the product of its address marginal and the singlet projector. Negative controls violate the color constraint and are detected. Checks overlap and are not independent physical observations.

The exact v0.8.2 manuscript and matching companion package are identified by https://doi.org/10.5281/zenodo.23497517. RELEASE.json binds the PDF, scripts, source manifest and reference outputs by SHA-256. The scientific scripts, eight frozen original sources and reference outputs are unchanged from the v0.6 package (DOI 10.5281/zenodo.23496836). SOURCE_SHA256.json is checked before execution. The upstream source is pinned to commit 9d9daf21154e46e55b9fd0fd461c7c070156ab94.

```
python3 -m venv .venv
.venv/bin/python -m pip install numpy==2.3.5 scipy==1.17.0 sympy==1.14.0
.venv/bin/python audit.py
.venv/bin/python dependency_audit.py
```

Recorded Python: 3.12.14. Reruns overwrite their result JSON files; preserve reference bytes for release-hash checks. Different runtime metadata or numerical libraries can change output hashes. Absolute fraction/share tolerances are 2 x 10^-12, historical mean-count tolerance 2 x 10^-10, matrix tolerance 10^-12 and numerical kernel threshold 10^-10; symbolic tests use exact zero. Known inputs and reproduced values remain legitimate model-construction and consistency results.

Conclusion

The frozen particle/component replay is reproducible and order dependent. Its supplied color test preserves the declared singlet but consumes no arithmetic state, and its exact three-factor singlet sector cannot retain address distinctions. These results specify the information that must be retained when extending this particle/component interface. This is a source-specific reproducibility note and conditional model-development result; general physical completion remains outside its claim.

Sources and established mathematical background

[1] W. Choi and J. Choi, frozen WRRA Stage-3 particle/component replay and Stage-2 handoff, repository github.com/Wonsik-Choi-janefather/wrra-m-0.1, commit 9d9daf21154e46e55b9fd0fd461c7c070156ab94. Exploratory source DOI: 10.5281/zenodo.23128482. These are computational provenance, not external experimental replication.

[2] D. Tong, The Standard Model, chapter 3, The Strong Force, University of Cambridge lecture notes, equations (3.4)-(3.6) and (3.95). https://davidtong.org/pdfs/teaching/standard-model/standardmodel3.pdf. Standard SU(3) and epsilon-singlet background.